Struct-X: Physics-Aware Event Vision for Robust Non-Contact Structural Vibration Monitoring

A Literature Survey, Gap Analysis and Preliminary Research Plan

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Abstract. Non-contact vibration measurement with an event camera is technically attractive: microsecond timestamps, ~140 dB dynamic range and negligible motion blur suit high-frequency structural response far better than contact or frame-based sensing. In practice, however, an event stream is not a picture of the structure but the residual of a sensor whose output depends on illumination, contrast threshold and bias currents, so the same physical motion is measured differently under different settings. This report surveys three methodological families that between them cover the pieces such a system needs: event-frame geometric deformation estimation, physics-guided learning for structural dynamics, and spatiotemporal Transformers with attention factorised over space and time. We find that the three have never been joined, that no reviewed work reports sensor-shift degradation as a first-class metric, and that spatiotemporal Transformers need more data than a 30-recording beam dataset can supply. We formalise the gap, propose a physics-aware event-vision pipeline with a reduced-order single-degree-of-freedom consistency term, and pre-register a falsification criterion: if the reduction in robustness gap does not exceed trial-to-trial variance, we report the physics term as not identifiable at this scale.

Index Terms. *event cameras, non-contact vibration measurement, structural health monitoring, physics-informed neural networks, spatiotemporal Transformers, domain shift, robustness.*

I. Introduction

A. Background

Vibration is the observable that structural health monitoring (SHM) has historically rested on: damage changes stiffness, mass and damping, and those changes leave detectable fingerprints in natural frequencies, mode shapes and acceleration responses. The instrumentation, however, is awkward. Accelerometers must be bonded to the member being watched, altering the very dynamics they measure. A laser Doppler vibrometer (LDV) avoids contact but needs a clear line of sight and a head held still while the target moves: workable on a laboratory bench, much less so on a bridge or a rotor. Cameras have therefore been pursued for years as a cheap, scalable, non-contact substitute, and event cameras have sharpened the idea considerably.

An event camera, or dynamic vision sensor, abandons fixed-rate sampling. Each pixel asynchronously reports when the log intensity at its own location changes by more than a contrast threshold, emitting an event $e = (x, y, t, p)$ with polarity $p \in \{-1, +1\}$ [1]. The consequences are exactly those vibration measurement wants: microsecond timestamps, roughly 140 dB of dynamic range against about 60 dB for conventional CMOS, per-pixel bandwidth on the order of kilohertz, and little motion blur [1], [2]. For a beam oscillating at tens of hertz to a few kilohertz, a conventional camera either blurs the motion or demands lighting and bandwidth a field deployment cannot carry. This is not a marginal advantage.

B. Motivation

There is a catch, and it is why this project exists. An event stream is not an image of the world; it is the residual of a sensor with a handful of user-set knobs. Illumination level and spectral content determine how many events a moving edge generates. The contrast threshold C sets how large a brightness step a pixel must see before it speaks, and threshold mismatch across pixels is itself a noise source: Gallego et al. report the effective threshold as a random variable distributed approximately normally about the nominal setting, with a spread of 2–4% temporal contrast [1]. Refractory periods, bias currents and bandwidth limits shape the rest of the noise floor. Change any of these and the sensor produces a *different measurement of the same physical motion*. The domain shift in this project lives in the sensor, not the structure, and it is present in every recording we will ever take.

Meanwhile the SHM literature has spent the last several years walking in the opposite direction. It has learned to keep models usable when the environment changes and the damage class is unknown, via open-set domain adaptation [8]; to inject governing equations into the training loss so that data-hungry networks still behave when data is scarce and noisy [6], [7]; and that video models built purely from self-attention need their attention factorised over space and time to be trainable at all [9]. What we could not find, after searching the families described below, is any work joining these threads to the event camera. That is the hole this project intends to fill.

II. Research Problem

Let E_s denote the event stream recorded under sensing configuration s (illumination, contrast threshold, bias setting, sensor variant), and $q(t)$ the true vibration response measured by a co-located LDV, which we treat as ground truth. We seek a mapping $f_\theta : E_s \rightarrow q(t)$ trained with a physics-aware objective, satisfying three requirements:

- **P1. Accuracy.** On held-out *recordings* from the configuration seen in training, $f_\theta(E_{s_0})$ should approximate $q(t)$ in waveform RMSE, MAE and correlation, and recover the dominant frequency f_n.
- **P2. Stability.** The *robustness gap* $\Delta = M(\text{shifted}) - M(\text{matched})$ should be small, where M is waveform RMSE or relative frequency error. This is what makes the approach a *sensing* method rather than a one-configuration demo.
- **P3. Feasibility.** The mapping must be learnable from 30 independent beam recordings [10], which forces windowed training data with trial-level splits and rules out any large model trained from scratch.

Two scope decisions follow. First, this is a study of vibration *estimation* and sensing robustness, not damage classification: the XJTU-DV Beam subset provides no damage labels, and we decline to claim what the data cannot support. Second, our regression target is a continuous waveform rather than a categorical label, which changes what "robustness" has to mean (Section VIII).

III. Objectives of the Literature Survey

The survey answers five questions in sequence, following the read → categorise → compare → critique → gap → direction chain rather than summarising papers one at a time.

1. **O1.** Establish what an event camera physically measures, and which of its parameters are set by the user rather than by the scene. This decides whether sensor-induced domain shift is real or theoretical.

2. **O2.** Identify pipelines that recover deformation or displacement from event data, recording what each assumes (brightness constancy, spatial continuity, co-registered frames) and what supervision it needs.
3. **O3.** Identify how physics is injected into deep learning for structural dynamics, recording which physical form the constraint takes and where the required parameters come from.
4. **O4.** Identify spatiotemporal architectures suited to event-derived tensors, with attention to their behaviour in low-data regimes.
5. **O5.** Determine how the literature measures, or fails to measure, degradation under changed sensing or environmental conditions for a continuous-valued target.

IV. Literature Search Methodology

Six sources were queried: IEEE Xplore, ScienceDirect (including *Mechanical Systems and Signal Processing* and *Engineering Structures*), SpringerLink, Wiley Online Library, the ACM Digital Library and arXiv; the dataset record was retrieved separately from Zenodo. Searching ran in three passes: a broad keyword pass, a backward-citation pass from the two anchor surveys [1], [5], and a forward-citation pass from the XJTU-DV dataset paper [10]. Keyword groups and their yields are in Table 1.

Table 1. Search protocol. 17 candidates retrieved across three passes; 12 retained after two-stage screening (5 core, 7 supporting).

Concept group	Keywords	Retained
Sensor	event camera; dynamic vision sensor; contrast threshold; illumination; noise	3
Task	non-contact vibration measurement; visual deformation measurement; beam	2
Method	physics-informed; physics-guided; neural network; structural dynamics	4
Architecture	spatiotemporal transformer; space-time attention; divided attention	2
Robustness	domain adaptation; open set; out-of-distribution	1

Records were excluded if they addressed only frame-based sensing, if the physical system was not a vibrating structure, or if the reported task was classification where our target is a waveform. Priority followed the course brief: recent journal papers, top-tier conference papers, foundational work, systematic reviews, then work directly on non-contact vibration sensing. All twelve retained sources are listed in Section XII.

V. Detailed Review of Existing Literature

The retained work clusters into three methodological families, reviewed here on their own terms before comparison.

A. Approach 1: Geometric, largely training-free event-frame deformation measurement

The most mature line of work relevant to us measures deformation directly from event data with an explicit motion model rather than a learned one. Wu et al. [4] fuse an event stream (temporally dense, spatially sparse, cheap) with a frame stream (spatially dense, low-noise) and solve for a dense displacement field. Their *Affine Invariant Simplicial* (AIS) framework partitions that field into low-parameter sub-regions and linearises each via simplicial motion parameterisation, exploiting the reproduction property $I[f] = f$ for affine $f(X) = AX + b$ to suppress the ambiguity sparse, noisy events introduce. Motion is recovered by optimising an event contrast-maximisation objective jointly with image cross-correlation, coarse-to-fine, with a neighborhood-greedy scheme letting well-converged sub-regions guide their neighbours. On a purpose-built benchmark of 120+ aligned event/frame sequences with high-speed-video ground truth, the method reaches a 65.7% survival rate beyond 100 px displacement (1.6× the previous state of the art) using 18.9% of the storage and processing budget of high-speed video.

Two features matter to us beyond those numbers. The method is entirely training-free, so there is no train/test split to leak and no data-scarcity problem. And its authors are explicit that no suitable event dataset existed, that building one is costly, that structural health monitoring is future work, and that the method leans on spatial continuity and brightness constancy, with pseudo-strains under topological change and degraded accuracy under specular reflection [4]. The wider sensor literature supplies the rest of this family: the representation taxonomy of Gallego et al. [1] (event frames and time surfaces retaining one timestamp per pixel; grid-like voxel tensors that drop into conventional CNN pipelines; learned representations) and the simulator conventions of Gehrig and Scaramuzza [2], which let a method be stress-tested against camera parameter changes without recapturing hardware.

B. Approach 2: Physics-informed and physics-guided learning for structural dynamics

This family treats physics as a training signal rather than an inference procedure. Raissi et al. [5] define the general form: a network is trained on a data-fidelity term while simultaneously satisfying the residual of a governing nonlinear PDE, in continuous-time and discrete-time variants, the latter admitting implicit Runge–Kutta schemes of arbitrary order. The framework is data-efficient precisely because the physics term constrains the hypothesis set; Karniadakis et al. [12] later generalise it and document where it fails: training can fail for reasons unconnected to the physics, and poorly chosen residuals actively hurt.

The structural instantiation closest to Struct-X is Wang et al.'s PhyResNet [6]. A ResNet encoder maps acceleration responses to low-dimensional structural states (stiffness reductions); a physics-based decoder (an updated finite element model solved by Newmark-β) reconstructs the response from those states. The equation of motion $M\ddot{x} + C\dot{x} + Kx = F(t)$ enters as one loss term, a second regresses the states against ground truth. The numbers establish the regime in which physics helps. On a simulated simply supported beam (10 elements) with 1–3% modelling error and 10%/20% Gaussian measurement noise, trained on 5% of the data, PhyResNet improves R-value by 13.1% over the pure ResNet; at 10% noise R rises from 0.789 to 0.870, and at 20% noise it holds 0.808; with modelling error alone, MSE falls from 2.0×10^{-4} to 1.6×10^{-4} and R from 0.805 to 0.850. A seven-storey frame experiment reproduces the trend, and the authors are candid that physics-informed learning in SHM remains early-stage.

Closest of all to our setting is the work reconstructing structural displacement from *both* accelerometer and camera measurements under a physics-informed objective [7]. That is structurally the same problem Struct-X faces, fusing a non-contact visual measurement with a physical response and letting the equation of motion regularise the fit. Its weakness is that the camera modality is a conventional frame camera, inheriting frame-camera failure modes under fast, small-amplitude motion, and its physics residual is written on the accelerometer channel rather than a high-bandwidth optical one.

C. Approach 3: Spatiotemporal Transformers and robustness-aware SHM

This family supplies the architecture. Bertasius et al. [9] remove convolution entirely and build video classification from self-attention over frame-level patches, comparing several attention schemes. Their operationally important finding: *divided attention*, applying temporal and spatial attention separately within each block, gives the best accuracy among the choices considered, whereas joint space-time attention is not merely worse but overflows GPU memory at the resolutions they target. TimeSformer

sets the best reported Kinetics-400 accuracy (80.7% top-1, large variant), trains faster than 3D convolutions, and handles clips over a minute long.

The same paper contains a warning we have taken seriously. Bertasius et al. attribute the convolutional bias (local connectivity and translation equivariance) as genuinely *beneficial on small training sets*, and argue Transformers pay off in big-data regimes where less inductive bias is needed; their spatial resolution ablation shows that accuracy gains from finer spatiotemporal partitioning require proportionally more pretraining. On robustness, Zhou and Lai [8] confront shift for classification, combining open-set domain adaptation with varying environmental conditions and unknown damage classes. The transferable idea is not their mechanism but their insistence that train and test conditions differ *by construction* rather than by random split.

VI. Comparative Analysis

Table 2 places the three families side by side on the axes that decide whether a method can be adapted to Struct-X. The last three rows matter most: none reports degradation under a changed sensing configuration, and only one needs no data at all, at the cost of a co-registered frame stream and an assumption of brightness constancy.

Table 2. Comparative analysis of the three approach families against requirements P1–P3 of Section II. Outputs are, respectively, a dense per-pixel displacement field, a damage index / reconstructed response, and a class label; none of the three emits a calibrated confidence.

Dimension	A. Geometric event-frame VDM [1], [2], [4]	B. Physics-guided structural DL [5]–[7], [12]	C. Spatiotemporal Transformer + robust SHM [8], [9]
Input representation	Raw events + co-registered frames; event frames, time surfaces, voxel grids	1-D sensor series (acceleration, displacement), sometimes FEM-derived features	Stacked voxel-grid / frame-level patches
Supervision	None (training-free optimisation); ground truth used only for evaluation	Supervised on damage indices or response traces; low-data tolerance is the headline result	Supervised, large-scale video labels; pretraining strongly recommended
Form of physics	Implicit: linearised solid-elastic / simplicial motion prior inside the estimator	Explicit PDE residual in the loss; the decoder may *be* an FEM (Newmark-β)	None
Needs an FE model?	No	**Yes** (or equivalent identified parameters)	No
Data regime	Any (per-sequence optimisation)	Modest; gains largest at 5–10% of data	Large; degrades with small samples [9]
Handles sensing shift?	Not evaluated	Not evaluated (noise / modelling error only)	Yes, by design [8], but categorical targets
Meets P1 (accuracy)?	Yes, for displacement	Yes, for damage index	Unproven for waveforms
Meets P2 (stability)?	Unknown	Partial (noise only)	Partial, wrong target type
Meets P3 (30 recordings)?	Yes	Yes	Only if pretrained

VII. Critical Analysis

Approach 1 buys robustness with assumptions we cannot accept. The AIS framework is elegant precisely because it refuses to learn, and its authors are honest that it leans on spatial continuity and brightness constancy. Both fail exactly where we need them. Structural vibration at tens to thousands of hertz produces displacements small in amplitude relative to pixel pitch and fast relative to any achievable frame rate; brightness constancy is violated by every lighting change in the shifted conditions, and spatial continuity is violated wherever the beam edge is the only thing generating events. The method also needs a synchronised, co-registered frame stream, and returns a field by per-sequence optimisation: no learned prior to amortise across 30 trials, and no calibrated uncertainty on the output.

Approach 2 has the right inductive bias pointed the wrong way. Every instantiation we found is formulated on *point* sensors with a known structural model. PhyResNet's decoder literally is a converged finite element model solved by Newmark-β, and the physics term is meaningful only because M, C, K are known. Struct-X inverts this: we have rich imagery and *no* FE model of the beam, and our ground truth is an LDV trace, already a physical measurement of the response. Borrowing PhyResNet's loss verbatim would require us to build and converge an FE model we do not have, importing exactly the modelling-error problems that paper measures. The honest adaptation is a *reduced-order* single-degree-of-freedom residual with m, c, k identified from a single training trial, which immediately creates the failure mode of Section IX.

Approach 3 is documented to be wrong at our data scale. We read the TimeSformer results as a constraint, not an endorsement. Divided attention is right on principle, but the same ablation that selects it shows accuracy degrading toward the data-driven baseline as pretraining data thins, and 30 recordings is far below the regime where any from-scratch video Transformer is defensible. Any Struct-X encoder must therefore be pretrained or deliberately lightweight, and we should report the CNN encoder too rather than quietly presenting the Transformer as the main model.

The cross-cutting omission is the metric. No reviewed paper reports degradation under a changed *sensing* configuration as a headline number. Approaches 1 and 2 report accuracy under measurement noise and FEM modelling error, a different perturbation: noise is additive and zero-mean, whereas a change in illumination or contrast threshold is a *multiplicative, structured* change in the event statistics that moves signal and noise together. Approach 3 confronts shift, but for categorical labels, where robustness is a confusion-matrix property; for a continuous waveform the equivalent notion, namely how far the reconstructed trace and recovered frequency move, has no accepted definition here. Establishing that definition is itself a contribution, and we propose it in Section VIII.

A methodological risk we should name. Almost all of this literature splits at the sample or window level. With sliding windows drawn from continuous recordings, neighbouring windows overlap in time and share structure response, so a random window split leaks test data into training and inflates every number. We will split by complete recording and report mean ± standard deviation across independent trials, never a single aggregate figure.

VIII. Research Gaps to Be Addressed

1. **G1. No event-vibration pipeline exists.** Event-based visual deformation measurement [4] is the closest artefact and positions itself short of SHM; no reviewed work predicts an LDV-referenced waveform from events.
2. **G2. Physics and events have never been combined.** Physics-guided structural networks consume point-sensor series [6], [7]; event pipelines are training-free or purely data-driven [4]. The two literatures do not cite each other.
3. **G3. Sensing-condition shift is unmeasured.** No reviewed study reports a robustness gap under changed illumination, contrast threshold or sensor configuration. The shift event cameras are *most* exposed to is the one *least* characterised.
4. **G4. No waveform-domain definition of robustness.** Reviewed robustness metrics are classification metrics [8] or additive-noise stress tests [6]; there is no continuous-target analogue for a measured physical response.
5. **G5. Physics priors are unidentified under scarce data.** In [6] the physical parameters are given. Where they must be esti-

mated from the same scarce data the network learns from, the residual can be driven to zero by the degenerate solution $\hat{q} \to 0$, and we found no treatment of this in the structural literature.

IX. Proposed Research Direction

We propose Struct-X: a physics-aware spatiotemporal event-vision pipeline trained on the XJTU-DV Beam subset [10] with LDV ground truth, and evaluated primarily on how little performance moves when the sensing configuration changes. Figure 1 shows the pipeline; Section X states the falsification criteria.

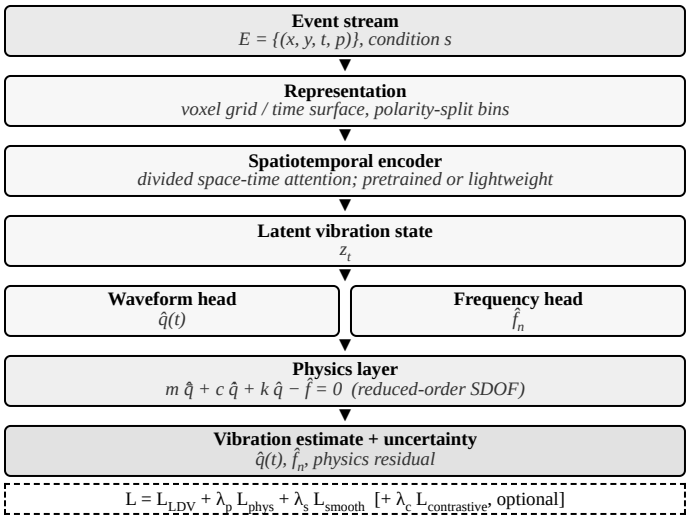


Fig. 1. Struct-X pipeline. Events are encoded into a spatiotemporal tensor, decoded into a vibration waveform and a dominant-frequency estimate, and constrained by a reduced-order single-degree-of-freedom residual. The contrastive term is an optional extension, not part of the core model.

Physics layer. For free vibration we adopt the damped single-mode model $m\,\ddot{q}(t) + c\,\dot{q}(t) + k\,q(t) = 0$, with natural frequency $f_n = (1/2\pi)\sqrt{(k/m)}$. The parameters are identified once by modal fitting on a *single* training trial and then frozen, so the physics term never sees test data. To defeat the degenerate collapse of G5, the residual is normalised by $\|\hat{q}\|$ so that shrinking the prediction toward zero does not reduce the loss; we also monitor the ratio of the two gradient contributions to confirm the data term still dominates early in training.

Experimental design. The central experiment is a leave-one-condition-out grid: train on condition *A*, then test on (i) held-out trials of *A* (the matched upper bound), (ii) low → high illumination, (iii) a different camera configuration, and (iv) a combined shift. Baselines are an FFT/feature-extraction pipeline, a CNN on the same representations, and a Transformer without the physics term. Metrics are waveform RMSE/MAE and correlation against LDV, relative frequency error $|\hat{f}_n - f_n|/f_n$, the normalised physics residual, and the robustness gap Δ of P2, all as mean ± standard deviation across trials.

X. Preliminary Research Questions

1. **RQ1 (feasibility).** Within one acquisition condition, to what waveform RMSE, correlation and relative frequency error can a spatiotemporal event model reconstruct the LDV-measured response on held-out recordings, and how much of that is attributable to the physics term rather than the architecture?
2. **RQ2 (mechanism).** Which constraint carries the benefit under scarcity (the SDOF equation residual, a temporal smoothness penalty on $\hat{q}$, or their combination) and does the benefit survive once m, c, k are identified from data rather than given?
3. **RQ3 (the actual claim).** Does the physics-aware model reduce the robustness gap Δ under illumination and contrast-threshold shift relative to each baseline, and is that reduction due to the physics term or merely to regularisation?
4. **RQ4 (which output is stable).** Is the dominant-frequency estimate more shift-robust than waveform RMSE? We expect so, since a spectral peak is a coarser statistic than a pointwise trace, but this is an assumption to test rather than to build on.
5. **RQ5 (falsification criterion, pre-registered).** If the reduction in Δ attributable to the physics term is smaller than the trial-to-trial standard deviation of Δ, we conclude the physics constraint is *not identifiable* at this data scale and report that as the outcome, negative result included. We will likewise report whether the physics term degrades the matched-condition result, since a regulariser that helps under shift while hurting in-distribution is a materially different finding.

XI. Conclusion

Non-contact vibration measurement with an event camera is technically well-motivated and, on the evidence gathered here, methodologically unclaimed. The three families surveyed each solve a third of the problem: geometric event-frame estimators recover deformation but assume brightness constancy and need frames; physics-guided structural networks supply the right inductive bias but demand a finite element model our setting does not have; spatiotemporal Transformers supply the right architecture but underperform when data is scarce. What is missing, and what this project proposes to test, is the joint claim: that a physics-aware spatiotemporal model can estimate LDV-referenced vibration from events *and* degrade less when the sensing configuration changes, measurably above trial-to-trial noise on 30 recordings. This is a feasibility study on one physical system rather than a benchmark, and RQ5 commits us to reporting a negative result plainly if the evidence does not support the claim.

XII. References

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Note. Table 1 records are our own; numbers quoted in Sections V–VII are as reported by the cited sources. All splits in Section IX are at recording level, not window level.